

APPLICATIONS STATISTICAL ADJUSTMENT OF TIDAL GAUGE DATA

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Summary: In this paper we consider measurements that were made using tidal gauges in order to construct tide tables. However, during the collection process, the measurements were corrupted due to the influence of the external air temperature. This led to the reported values being incorrect. It is shown how additional experimental data can be used to obtain estimates of the true values. Confidence intervals for these unknown true values are constructed using the bootstrap.

1. Introduction

Many problems we encounter in practice are of the following general form. There is some physical quantity of interest to us and (fairly exact) measurements on this can be made on a regular basis. The measurement process is often automated using an electronic device. However, sometimes these measurements become corrupted due to the influence of some physical factors, for example severe weather conditions. Often it is not possible or too expensive to redo the measurements, but there is still a need to obtain some information from the corrupted data. The problem then is how these historical measurements can be adjusted for the corruption in order to provide

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some useful information on the quantity of interest. In some situations a priori information is available to do this whilst in other situations some further experimental measurements can be made in order to carry out such an adjustment. In this paper we discuss an example of the latter situation.

Tide tables are based on tide measurements made on a regular basis with tidal gauges. These measure tidal heights and schematically work as in Figure 1 below.

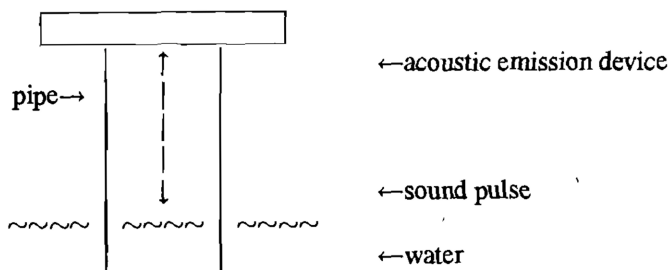


Figure 1. Schematics of a Tidal Gauge

In a very simplified form a tidal gauge works as follows:

A sound pulse is emitted from an acoustic device mounted at the top of a cylindrical pipe, open at the bottom, and immersed in the water. The time (t) taken for the pulse to travel down to the water and back again to the acoustic device is measured and recorded. Using a known speed of sound (s), a height (distance from the acoustic device to the water) is obtained from the well-known formula:

$$height = \frac{1}{2} s.t \quad (1)$$

The measured heights form the basis for calculating tidal heights (at a specific time, for a specific location) and thus tidal tables.

In practice, making such measurements with electronic equipment in the ocean often leads to problems due to the potentially adverse effect of sea water. However, other factors can also lead to corrupted data as we experienced.

It was discovered that some of the tidal values at high air temperatures were suspect. Investigation of this led to the following.

Before the measurement process is started, the gauge is calibrated at a known height of 1 metre, for which a speed of sound is obtained. This speed is then used in (1) in transforming the time measurements to heights and thus to tidal values. However, in this particular case, the gauge was not well insulated against temperature changes in the air, especially during hot days. This led to the temperature in the tube increasing during hot days, compared to the temperature at calibration, leading to the air being warmer at the top and cooler close to the water. Since the speed of sound is sensitive to temperature changes, this meant that an incorrect speed was used for some of the tidal calculations leading to the suspect values. The question asked was: What, if anything, could be done to salvage the data? How can the data be "adjusted" to compensate for the incorrect values used for the speed of sound?

In statistical terms the problem can be formulated as follows.

Estimate the unknown height D corresponding to an observed t_0 , the time taken for a sound pulse between emission and recording, at a given (fixed) external air temperature of T_0 .

In this paper we will discuss a possible solution to this problem. In the next section the adjustment procedure is discussed and in section 3 estimation of D , followed by confidence intervals for D in section 4. Section 5 is devoted to an example and some remarks are given in the last section.

2. Adjustment procedure

An essential aspect of an adjustment procedure is knowledge of the relationship between temperature (in the pipe) and depth within the pipe. This relationship can be obtained through experimentation and will be discussed below. However, before doing this, we first discuss the adjustment procedure. We do this in deterministic terms and will discuss the random aspect when discussing the experimentation.

Now, note that in air of constant temperature we have the usual relationship (1) between distance, speed and time, viz

$$d = \frac{1}{2} s \cdot t$$

or

$$t = 2d/s, \quad (2)$$

with

t = time for the sound pulse to travel down to the water and back to the emission device,

s = speed of sound,

d = height or depth (distance).

While formula (2) expresses the relationship between distance, time and a constant speed of sound, we have to allow for the fact that in our case the speed is not constant. It depends on the air temperature in the pipe, T , and it in turn depends, in our case on the depth, d . Thus

$$s = h(T) \quad (3)$$

and

$$T = g(d). \quad (4)$$

Writing $f = h \circ g$, the composition of h and g , this leads to

$$s = f(d). \quad (5)$$

Substituting into (2) gives

$$t = 2d/f(d). \quad (6)$$

Now, for a given depth D , divide the interval $[0, D]$ into k subintervals of lengths $\Delta_i, i = 1, 2, \dots, k$ with end points

$$0 = d_0 < d_1 < \dots < d_k = D.$$

Choose all Δ_i small enough (and thus k large enough) so we can assume a constant temperature in every subinterval $[d_{i-1}, d_i]$, and thus a constant speed of sound in each subinterval, say $s_i, i = 1, 2, \dots, k$.

Denote by t_i the time taken for a pulse to pass down and up through $[d_{i-1}, d_i]$. Then, since we assume constant speed of sound in each subinterval, we can use (6) to write

$$t_i = 2(d_i - d_{i-1})/f(d_i), i = 1, 2, \dots, k. \quad (7)$$

Adding these t_i gives the time it takes for a pulse to travel a distance D and back and with temperature changes governed by the function g . Denote this time by $t(D)$; substituting from (7) we then obtain

$$\begin{aligned} t(D) &\equiv \sum_{i=1}^k t_i \\ &= \sum_{i=1}^k \frac{2(d_i - d_{i-1})}{f(d_i)} = \sum_{i=1}^k \frac{2\Delta_i}{f(d_i)}. \end{aligned} \quad (8)$$

The right-hand side of (8) is a Riemann sum in which we can let $k \rightarrow \infty$ in such a way that

$$\max_{1 \leq i \leq k} \Delta_i \rightarrow 0,$$

leading then to the corresponding Riemann integral. The left-hand side remains $t(D)$, the time taken to travel down a distance D and back. In the

limit we can, in particular, assume constant speed of sound (over infinitesimal intervals) and thus (7) and (8) are applicable, giving

$$t(D) = \int_0^D \frac{2du}{f(u)}. \quad (9)$$

Thus, for given t_0 , we can find the unknown D_0 corresponding to this, as that value for which

$$t(D_0) = t_0,$$

i.e. solve D_0 from

$$\int_0^{D_0} \frac{2du}{f(u)} = t_0, \quad (10)$$

with $f = h \circ g$.

In order to apply this, we need the functions h and g . We now discuss this.

Consider h first. It is known (see e.g. Pierce, 1989) that the speed of sound in dry air is given by

$$s = h(T) = 332(1 + T/273)^{\frac{1}{2}}, \quad (11)$$

with T in degrees Celsius. This is for dry air, in humid air the following approximation holds

$$s_{wet} \simeq (1 + 0.016\eta)s_{dry},$$

where η is the fraction of water molecules in the air. This correction is typically less than 1.5 % since η rarely exceeds 0.07 (100 % humidity at 40 degrees Celsius, as given in e.g. Pierce, 1989). Since this correction is negligible, we did not take this into account and used (11).

A known form for g is not available and experimental data had to be collected to estimate it.

For a given air temperature, measure the temperature in the tube of the gauge at different depths, say $d_1, d_2, \dots, d_n$. This can be done by inserting thermometers through the tube at these depths. This experiment produces the

data

$$(d_1, T_1), (d_2, T_2), \dots, (d_n, T_n).$$

Assume an additive model, ie.

$$T = g(d) + e, \quad (12)$$

with g unknown and some random error e . Note that, although we do not indicate it in our notation, the model (and g in particular) is a function of the air temperature (see also Remark 5.1 below).

Since we have no knowledge of the form of g , we fitted a non-parametric function to the data to estimate g . We tried different procedures, all giving much the same results – we decided on the R-procedure `smooth.spline` (see the R-Library, Version 1.8.1, 2003). Denote the estimated function by $\hat{g}$. For this estimated $\hat{g}$ and a given t_0 , we can thus set up the equation

$$\int^D \frac{2du}{h(\hat{g}(u))} = t_0 \quad (13)$$

to obtain an estimator for D_0 , say $\hat{D}$. We solved this equation using the R-function `uniroot`.

Note that there is also some variation in the time measurements, from pulse to pulse, due to local movement of the sea. However, tidal gauges are usually deployed in very sheltered areas so that this variation is small. We did not take it into account due to its negligible effect on our results.

In the next section we show how to obtain a confidence interval for D_0 based on $\hat{D}$.

3. Confidence interval for D

In order to obtain a confidence interval for D , we use the well-known bootstrap procedure (see e.g. Efron and Tibshirani, 1993). We used four bootstrap

procedures from the R-package boot. These are the percentile, BCA, normal and basic methods. They all gave essentially the same results.

Since g is the only quantity that had to be estimated prior to estimating D , we apply the bootstrap to $\hat{g}$. This is done as follows.

Define the residuals

$$r_i = T_i - \hat{g}(d_i), \quad i = 1, 2, \dots, n. \quad (14)$$

From these take a sample with replacement say $r_i^*, i = 1, 2, \dots, n$, and construct the pseudo observations

$$T_i^* = \hat{g}(d_i) + r_i^*, \quad i = 1, 2, \dots, n. \quad (15)$$

Use (d_i, T_i^*) to find $\hat{g}^*$ (using the same non-parametric regression estimation procedure as before) and thus $\hat{D}^*$ from the equivalent of (13), viz

$$\int_0^D \frac{2du}{h(\hat{g}^*(u))} = t_0. \quad (16)$$

Repeat this B times to get

$$\hat{D}_1^*, \hat{D}_2^*, \dots, \hat{D}_B^*.$$

Denote the corresponding ordered values by

$$\hat{D}_{(1)}^* < \hat{D}_{(2)}^* < \dots < \hat{D}_{(B)}^*.$$

As a $(1 - 2\alpha)$ confidence interval for D , use

$$\left(\hat{D}_{(B\alpha)}^*, \hat{D}_{(B(1-\alpha))}^* \right). \quad (17)$$

In order to study the coverage probabilities of these intervals, we used a double bootstrap. Using this, we calibrated the intervals (i.e. adjusted the α in (17)) in order to obtain the desired coverage probabilities. The method used was as described by Efron and Tibshirani (1993), page 265. An alternative to the bootstrap is to use asymptotic theory. We remark on it in the final section.

4. An example

Consider the case where the speed of sound obtained from the calibration is 342.77 m/sec, corresponding to an air temperature in the tube of 18 degrees Celsius. Furthermore, we have $n = 11$ (the number of tube measurements) and we are interested in 3 values of (erroneous) depths, viz. $d = 2.2, 3.0$ and 3.5 metres.

We first need to obtain the measured time, i.e. the time value from which d was obtained through the formula (1). Now, for $d = 2.2$, we have

$$t_0 = \frac{2d}{s} = \frac{2(2.2)}{342.77} = 0.0128 \text{ sec.}$$

Similarly for $d = 3.0$, we get

$$t_0 = \frac{2(3.0)}{342.77} = 0.0175 \text{ sec}$$

and for $d = 3.5$, we get

$$t_0 = 0.0204 \text{ sec.}$$

In Figure 2 below we show the (d_i, T_i) data and estimated $\hat{g}$ function.

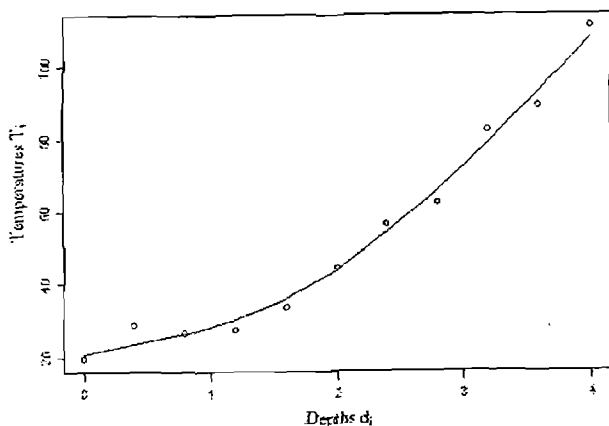


Figure 2. Data (d_i, T_i) and estimated $\hat{g}$

We applied the results of sections 2 and 3, using the R-subroutines `smooth.spline` and `unitroot`. The double bootstrap was used with 1000 replicates in each of the inner and outer bootstraps. Calibration of the confidence intervals led to some adjustment of the intervals, especially at the lower end. The following results were obtained for the 3 cases above

d	Estimate $\hat{D}$	95 % CI
2.2 m	2.24 m	(2.22, 2.25)
3.0 m	3.10 m	(3.06, 3.13)
3.5 m	3.65 m	(3.58, 3.69)

We observe that

- For $d = 2.2$ m we adjust to $\hat{D} = 2.24$ m, i.e. a 4 cm adjustment.
- For $d = 3.0$ m we adjust to $\hat{D} = 3.10$ m, a 10 cm adjustment.
- For $d = 3.5$ m we adjust to $\hat{D} = 3.65$ m, a 15 cm adjustment.

Note that we have fairly precise measurements and thus g is estimated with high precision. This is also reflected in the short confidence intervals for D given above.

5. Final remarks

- 5.1 Note that these results were for a particular choice of the external air temperature T_0 . For each choice of T_0 we have a different g and thus different estimates $\hat{g}$. The procedure is of course the same at other values of T_0 . Furthermore, it is quite plausible that g could vary from time to time, even at the same air temperature. Ideally one would want repeat data similar to what was used in this analysis. Unfortunately this was not available and we had to confine ourselves to the present data set.
- 5.2 A natural question is whether one can say anything about asymptotic theory as $n \rightarrow \infty$. The following gives some indication of this.

Writing

$$b(x) = \frac{2}{h(x)}, \quad (18)$$

we have $\hat{D}$ and D as the solutions of

$$\int_0^{\hat{D}} b(\hat{g}(u)) du = t_0 \quad (19)$$

and

$$\int_0^D b(g(u)) du = t_0, \quad (20)$$

respectively, for given t_0 .

Subtracting (19) and (20), we find

$$\begin{aligned} 0 &= \int_0^{\hat{D}} b(\hat{g}(u)) du - \int_0^D b(g(u)) du \\ &= \int_0^D (b(\hat{g}(u)) - b(g(u))) du + \int_D^{\hat{D}} b(g(u)) du \\ &\quad + \int_D^{\hat{D}} (b(\hat{g}(u)) - b(g(u))) du \\ &= (i) + (ii) + (iii). \end{aligned} \quad (21)$$

Note that we have conveniently taken $D < \hat{D}$; this of course need not be true, but since we are only arguing intuitively, it does not make any essential difference (except to complicate the discussion). The term (iii) is clearly of a smaller order than the first two terms and can thus be ignored asymptotically. To each of the first two terms in (21) apply a first-order Taylor expansion to get

$$\begin{aligned} 0 &= \int_0^D (\hat{g}(u) - g(u)) b'(g(u)) du \\ &\quad + (\hat{D} - D) b(g(D)) + \text{smaller order terms.} \end{aligned}$$

This gives

$$\hat{D} - D \doteq - (b(g(D)))^{-1} \int_0^D (\hat{g}(u) - g(u)) b'(g(u)) du. \quad (22)$$

This shows that the limiting behaviour of $\hat{D} - D$ is closely tied to that of

$\hat{g} - g$. Of course, the integral operator in (22) has a smoothing effect on the latter. In order to obtain the limiting distribution of $\hat{D} - D$ (suitably normalized), one can possibly use results on the weak convergence of kernel smoothers (see Hart, 1997, e.g. the proof of Theorem 6.2, p. 155) applied to $\hat{g} - g$, suitably normalized. Further work on this will be reported on in the future.

- 5.3 The problem can also be put in a general setting as follows. Consider $L(g, d)$, some functional of g and a function of d , some parameter of interest. The true value of d , say D , is given as the solution of

$$L(g, d) = 0. \quad (23)$$

However, g is typically unknown and has to be estimated, by say $\hat{g}$. (Note that g can be considered a functional nuisance parameter.) An estimator for D is then obtained by solving

$$L(\hat{g}, d) = 0. \quad (24)$$

Assume we can obtain data from a model

$$y = g(\underline{x}) + e, \quad (25)$$

for some set of covariates $\underline{x}$. For design points $\underline{x}_1, \dots, \underline{x}_n$ one then obtains data

$$(y_i, \underline{x}_i), \quad i = 1, 2, \dots, n,$$

from which $\hat{g}$ as a nonparametric regression estimator can be obtained. Using this in (24) and solving for d , gives $\hat{D}$. A bootstrap confidence interval can be obtained from this in a straightforward fashion as before. Furthermore, by writing

$$\begin{aligned} 0 &= L(\hat{g}, \hat{D}) - L(g, D) \\ &= (L(\hat{g}, D) - L(g, D)) + (L(g, \hat{D}) - L(g, D)) \\ &\quad + \left[(L(\hat{g}, \hat{D}) - L(\hat{g}, D)) - (L(g, \hat{D}) - L(g, D)) \right], \end{aligned} \quad (26)$$

ignoring the third term due to it being of a smaller order than the first two, developing the first two terms in first-order Taylor series, we obtain approximately

$$\hat{D} - D \simeq -L'(g, D)^{-1} L'_g(\hat{g} - g, D). \quad (27)$$

Here L' is the first derivative of L w.r.t. its second argument and L'_g

the first (functional) derivative of L w.r.t. its first argument. Note that L'_g is a linear functional in its first argument. The limiting distribution of $\hat{g} - g$ is thus intimately connected to that of $L'_g(\hat{g} - g, D)$. Furthermore, the limiting distribution for $\hat{D} - D$ (suitably normalized) then follows directly from the latter.

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